People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research

Houari Boumediene University of Science and Technology

Faculty of Computer Science

Data mining

Lab 4 Raport

Visual and Quantitative Comparison of Clustering Algorithms

4th year : *Software Engineering*

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I. Introduction

Data mining is an analytical approach aimed at extracting relevant and novel information from large datasets. Clustering, a fundamental technique in unsupervised data mining, enables grouping similar data without prior knowledge of categories.

This report presents **DMI**, an advanced interactive application developed in Python to visually and quantitatively compare different clustering algorithms. The application provides a comprehensive platform for data preprocessing, algorithm comparison, and result visualization.

II. Lab Objective

The main objective of this practical work is to:

- **Compare** multiple clustering algorithms (K-Means, K-Medoids, DBSCAN, Agglomerative, Diana)
- Implement classification algorithms (KNN, Decision Tree, Naive Bayes, SVM)
- **Visualize** clustering results interactively in 2D and 3D
- **Evaluate** algorithm performance quantitatively using standardized metrics
- **Provide** an ergonomic interface enabling data preprocessing and analysis
- **Support** multiple data formats and predefined datasets

III. Application Architecture

i. Technologies Used

- **Language:** Python 3.x
- **Graphical Interface:**
  - Tkinter (main GUI framework)
  - ttk (themed widgets)
- **Analysis Libraries:**
  - pandas: data manipulation
  - numpy: numerical computations
  - scikit-learn: clustering algorithms and metrics

  - scipy: statistical and hierarchical analysis
  - pyclustering: K-Medoids implementation

- **Visualization:**
  - matplotlib: 2D/3D graphics
  - FigureCanvasTkAgg: Tkinter integration
  - mpl_toolkits.mplot3d: 3D visualization support

ii. Modular Structure

The application is organized into five main tabs:

1. **Welcome**: Application introduction and quick start guide
2. **Preview**: Data loading, exploration, and initial statistics
3. **Preprocessing:** Data cleaning and transformation operations
4. **Clustering**: Algorithm selection, execution, and visualization
5. **Classification**: Supervised learning with multiple algorithms
6. **Results Tab**: Comprehensive analysis and summary statistics

IV. Implemented Features

i. Data Preview

Data Loading

Supported sources:

- Custom CSV files
- Predefined datasets (Iris,Wine,Blobs)

Data Preview

Data Head Tab:

- First 10 rows display in scrollable table
- All columns visible with horizontal scrolling
- Column headers with truncation for long names

Info & Stats Tab:

- **Dataset Info Panel:**

  - Column names
  - Data types (int64, float64, object)
  - Missing value counts per column

- **Statistics Panel:**
  - Minimum, Maximum values
  - Quartiles (Q1, Median, Q3)
  - Mean and Mode
  - Available only for numeric columns

ii. Preprocessing

1. Categorical Data Conversion:

- **One-Hot Encoding**: Creates binary columns for each category
- **Label Encoding**: Assigns numeric labels to categories
- Automatic detection of categorical columns
- Feedback on converted columns

2. Missing Values Handling:

- **Mean Imputation**: Replace with column average
- **Median Imputation**: Replace with column median
- **Mode Imputation**: Replace with most frequent value
- **Drop Columns**: Remove columns with >60% missing values
- Column-wise processing with detailed reporting

3. Duplicate Handling:

- Detection of duplicate rows
- User confirmation before removal
- Count of removed duplicates
- Data integrity preservation

4. Feature Scaling:

- **Min-Max Scaling**
- **Z-Score (Standard) Scaling**
- Applied to all numeric columns
- Essential for distance-based algorithms

Preprocessing Results Display:

- Real-time feedback in text widget
- Operation history tracking

- Scrollable results panel
- Clear separation between operations

iii. Clustering

Implemented Algorithms:

1. **K-Means**
2. **K-Medoids**
3. **Agglomerative (AGNES)**
4. **Diana**
5. **DBSCAN**

Analysis Tools

1. Elbow Method:

- Tests K values from 1 to 10
- Plots inertia vs. number of clusters
- Helps determine optimal K
- Interactive plot with grid

2. Dendrogram:

- Hierarchical tree visualization
- Uses Ward linkage method
- Samples up to 100 points for performance
- Shows cluster formation process

Results Export

Export Capabilities:

- Save cluster labels to CSV
- Preserves latest clustering result
- User-selected file location
- Default filename: cluster_labels.csv
- Error handling for file operations

iv. Classification

Implemented Algorithms:

1. **KNN (K-Nearest Neighbors)**
2. **Decision Tree**
3. **Naive Bayes** (Gaussian)
4. **SVM (Support Vector Machine)**

Classification Process

- The system stores the original (non-normalized) target values
- Prevents data leakage from preprocessing operations
- Checks for missing values in target column
- Ensures at least 2 classes present
- Detects continuous targets (>20 unique values) and rejects them
- 70% training, 30% testing (test_size=0.3)
- Random state = 42 for reproducibility
- Uses original target for training/testing
- Features: all numeric columns except target

Resultat Display:

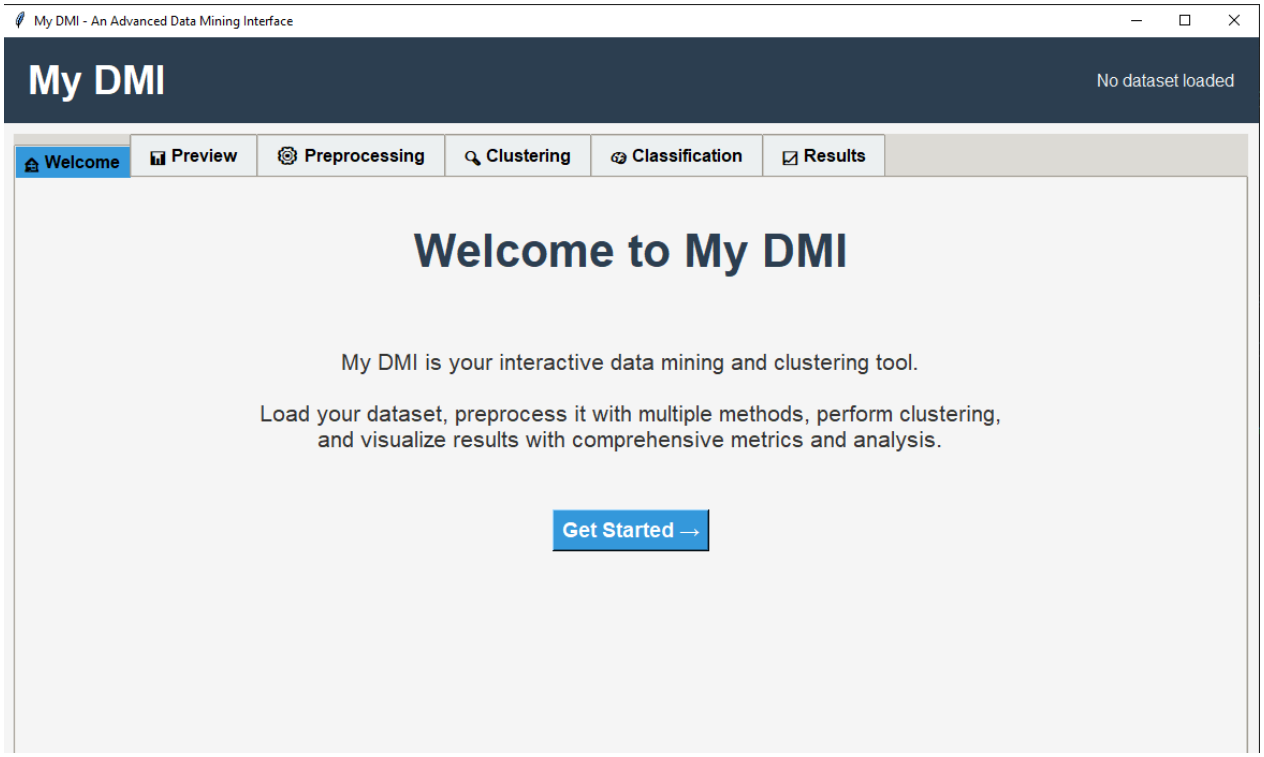
Classification Results Table:

- Algorithm name/Accuracy score /Target column used

Confusion Matrix Visualization

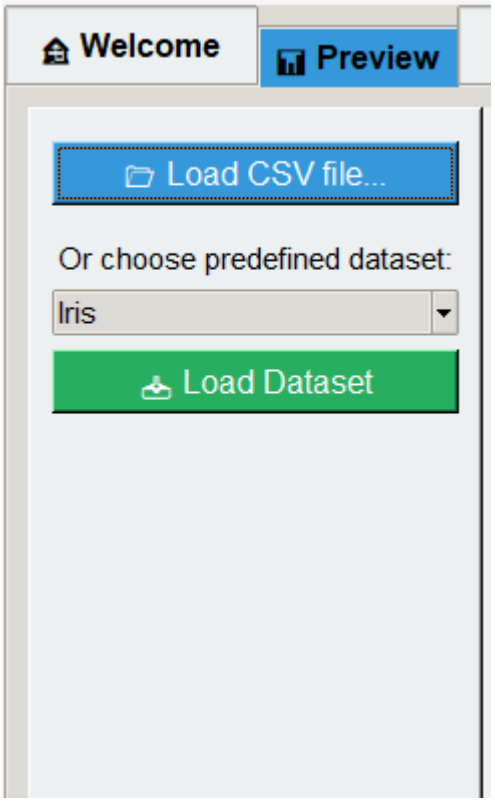
V. User Interface

Welcome Tab

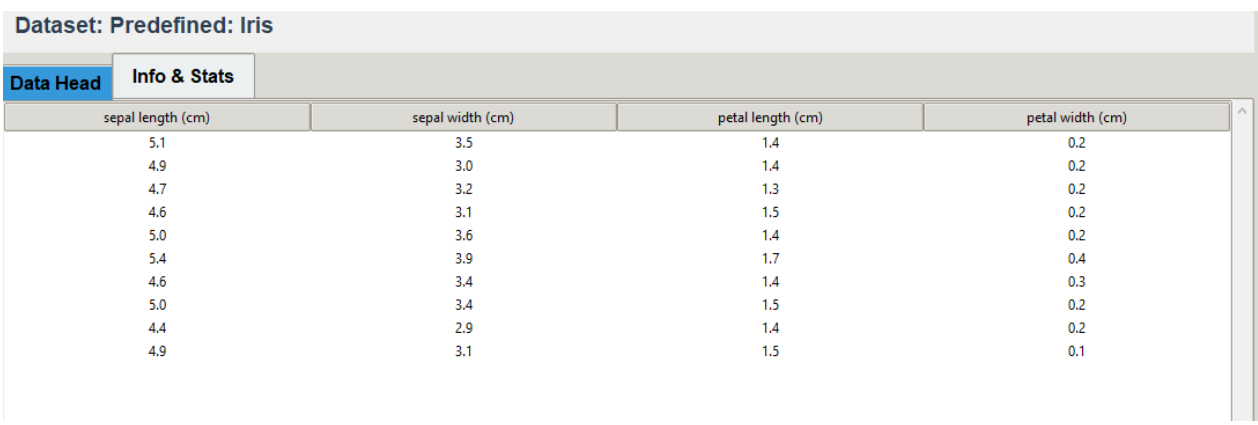


Preview Tab

Data Loading Panel

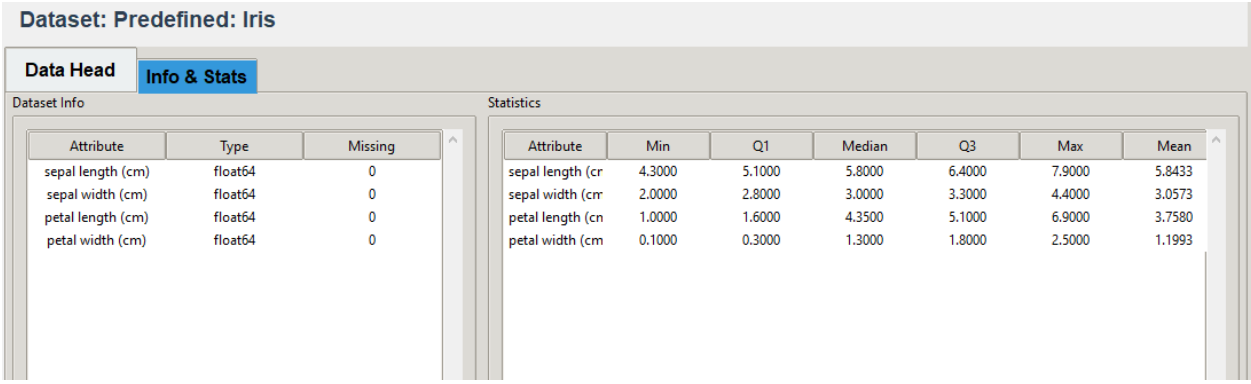


Data Head View



sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.0	3.6	1.4	0.2
5.4	3.9	1.7	0.4
4.6	3.4	1.4	0.3
5.0	3.4	1.5	0.2
4.4	2.9	1.4	0.2
4.9	3.1	1.5	0.1

Info & Stats View

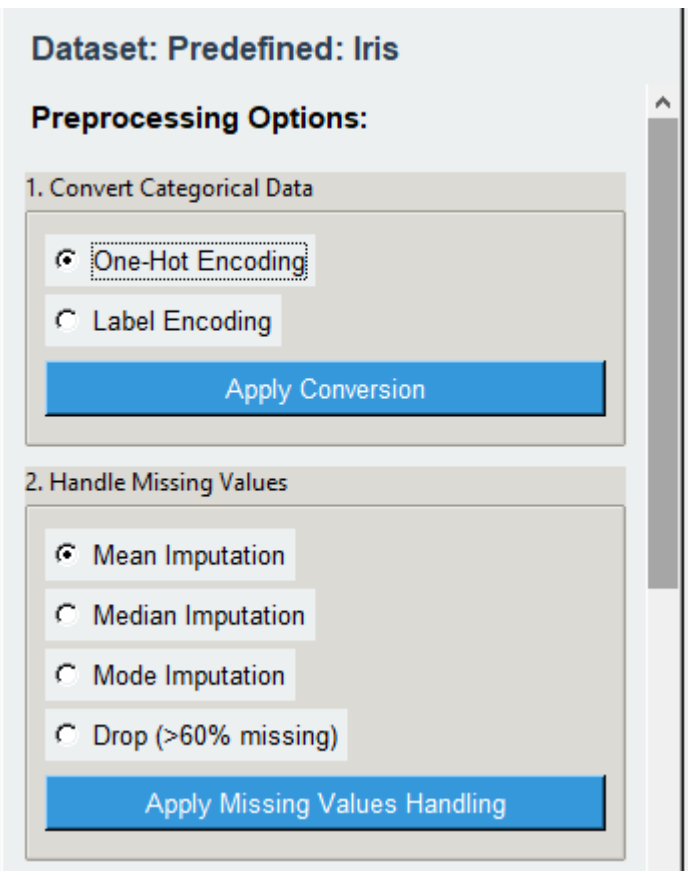


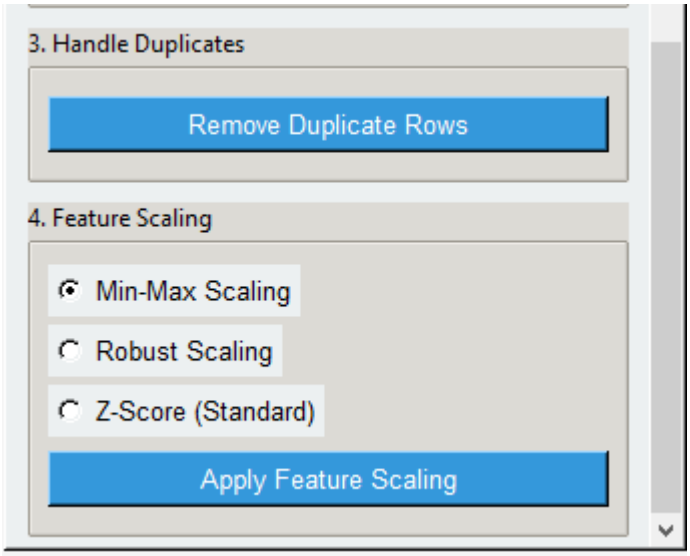
Attribute	Type	Missing
sepal length (cm)	float64	0
sepal width (cm)	float64	0
petal length (cm)	float64	0
petal width (cm)	float64	0

Attribute	Min	Q1	Median	Q3	Max	Mean
sepal length (cr	4.3000	5.1000	5.8000	6.4000	7.9000	5.8433
sepal width (cm	2.0000	2.8000	3.0000	3.3000	4.4000	3.0573
petal length (cn	1.0000	1.6000	4.3500	5.1000	6.9000	3.7580
petal width (cm	0.1000	0.3000	1.3000	1.8000	2.5000	1.1993

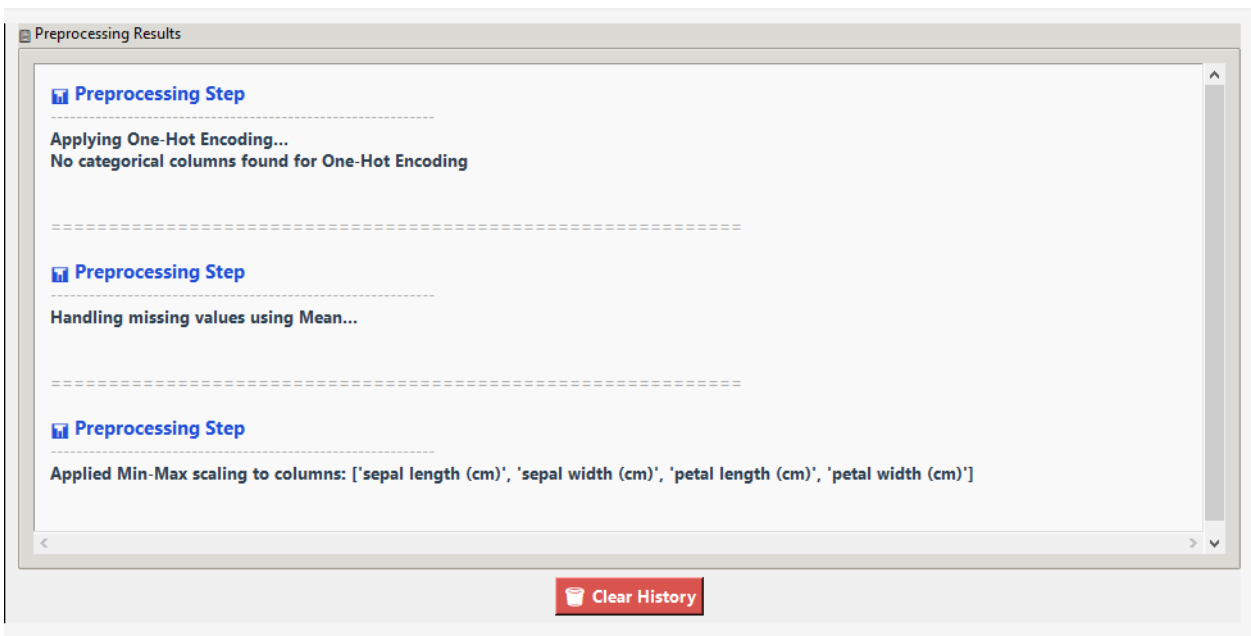
Preprocessing Tab

Preprocessing Controls



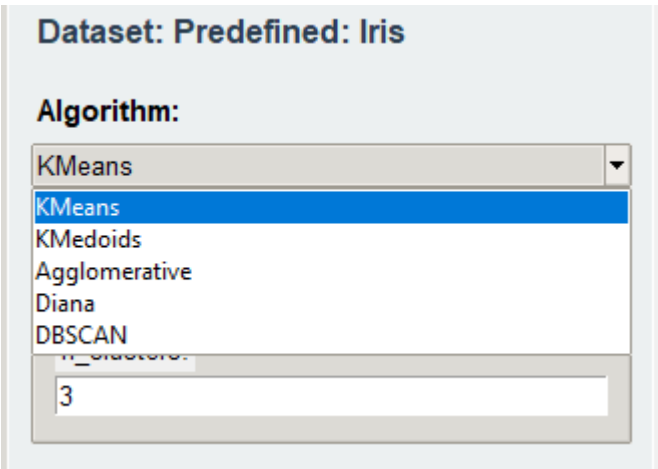


Preprocessing Results Display

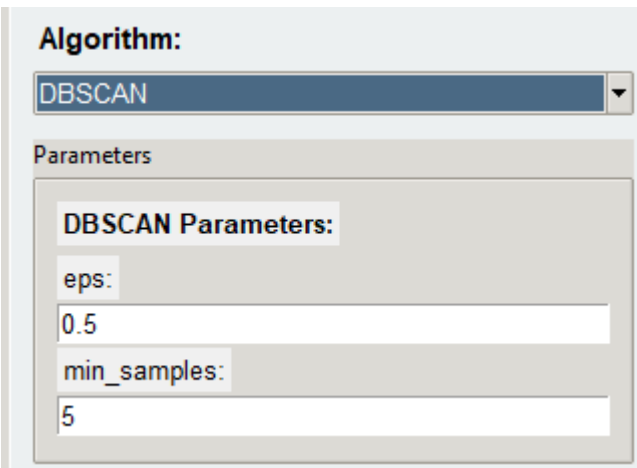


Clustering Tab

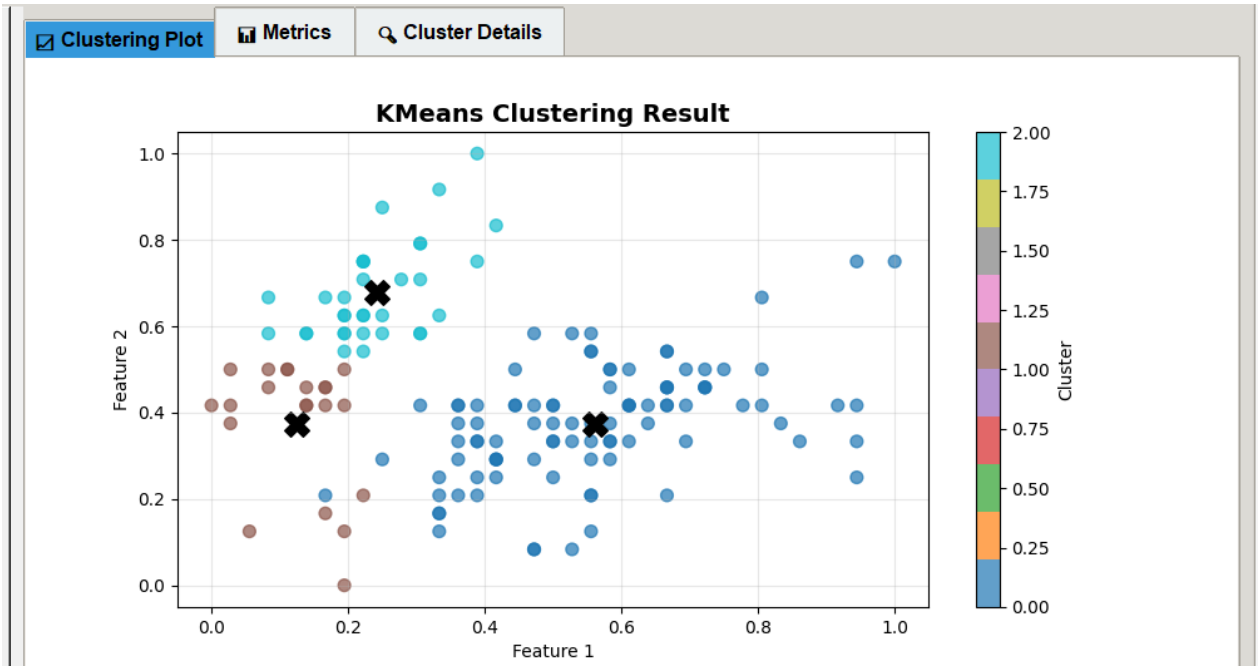
Algorithm Selection



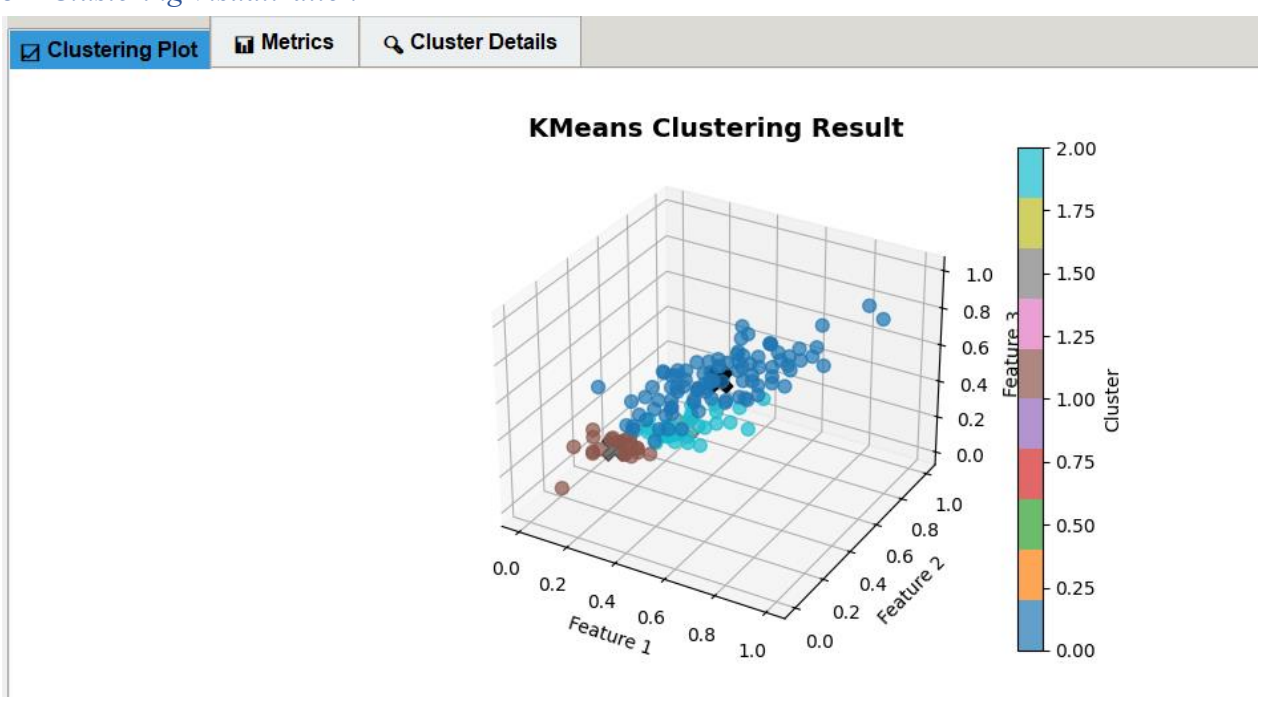
Example:



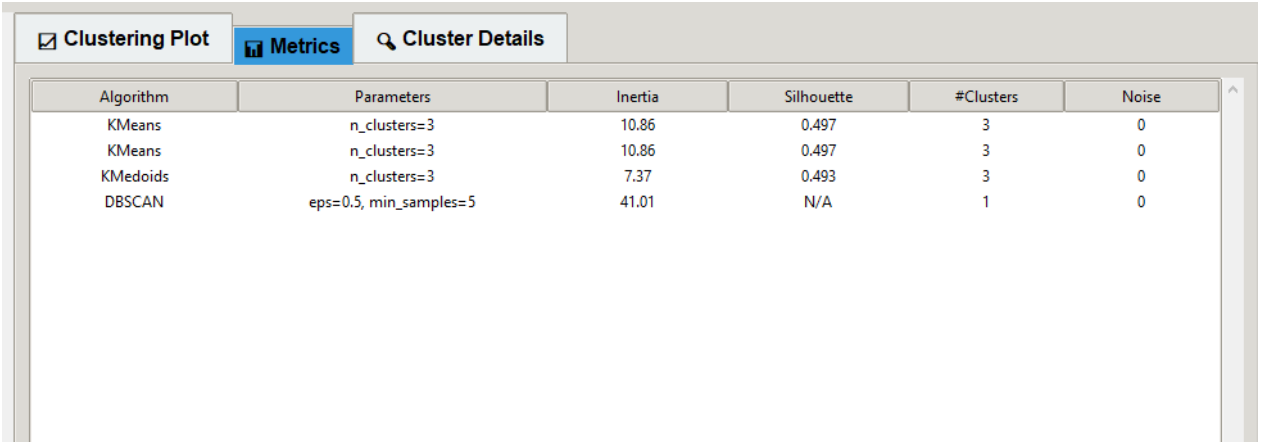
2D Clustering Visualization



3D Clustering Visualization



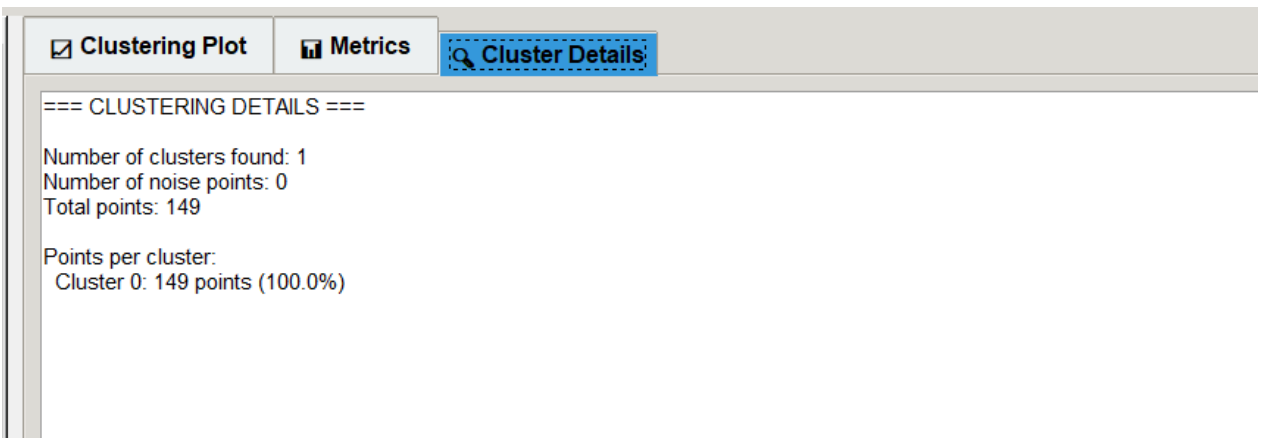
Metrics Table



Clustering Plot | Metrics | Cluster Details

Algorithm	Parameters	Inertia	Silhouette	#Clusters	Noise
KMeans	n_clusters=3	10.86	0.497	3	0
KMeans	n_clusters=3	10.86	0.497	3	0
KMedoids	n_clusters=3	7.37	0.493	3	0
DBSCAN	eps=0.5, min_samples=5	41.01	N/A	1	0

Latest Cluster Details



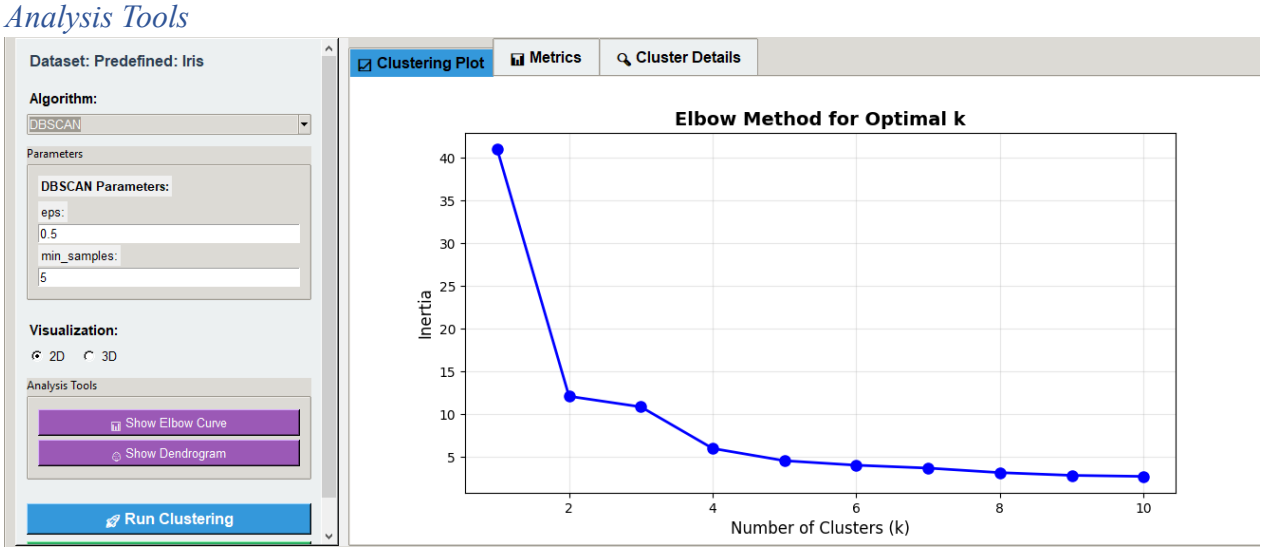
Clustering Plot | Metrics | Cluster Details

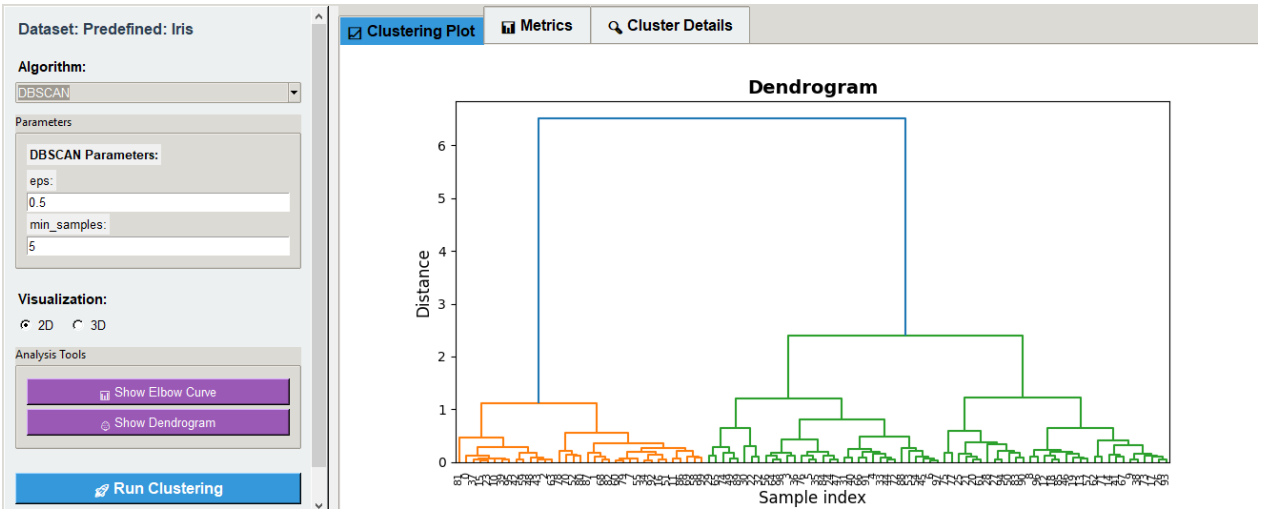
```
=== CLUSTERING DETAILS ===

Number of clusters found: 1
Number of noise points: 0
Total points: 149

Points per cluster:
  Cluster 0: 149 points (100.0%)
```

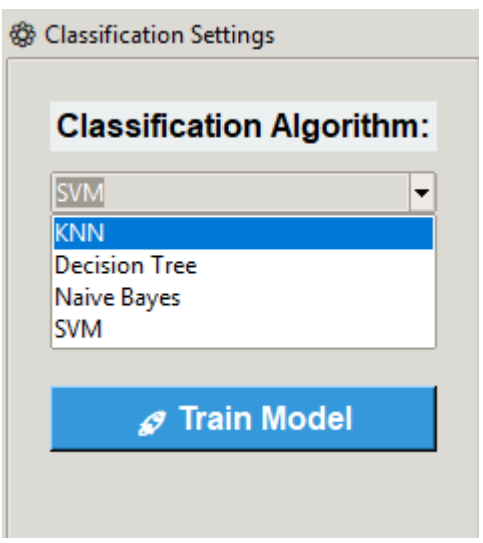
Analysis Tools



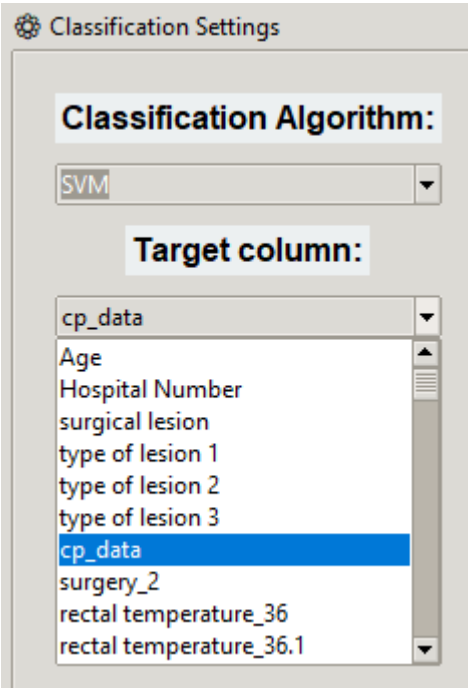


Classification Tab

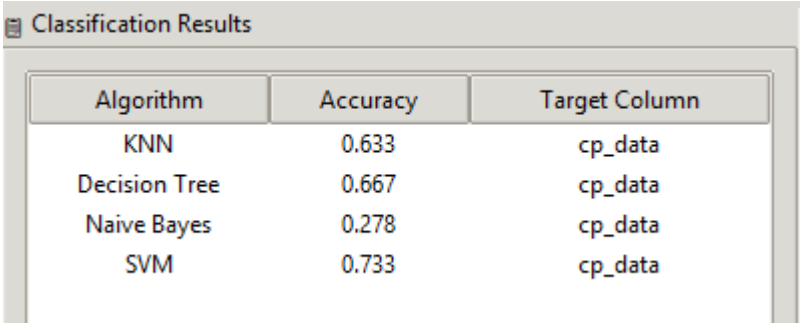
Algorithm Selection



Target column Selection



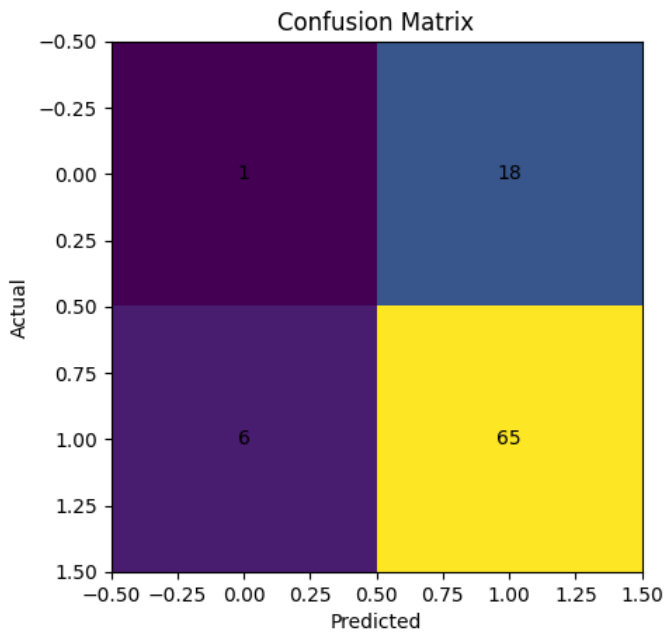
Classification results



Classification Results

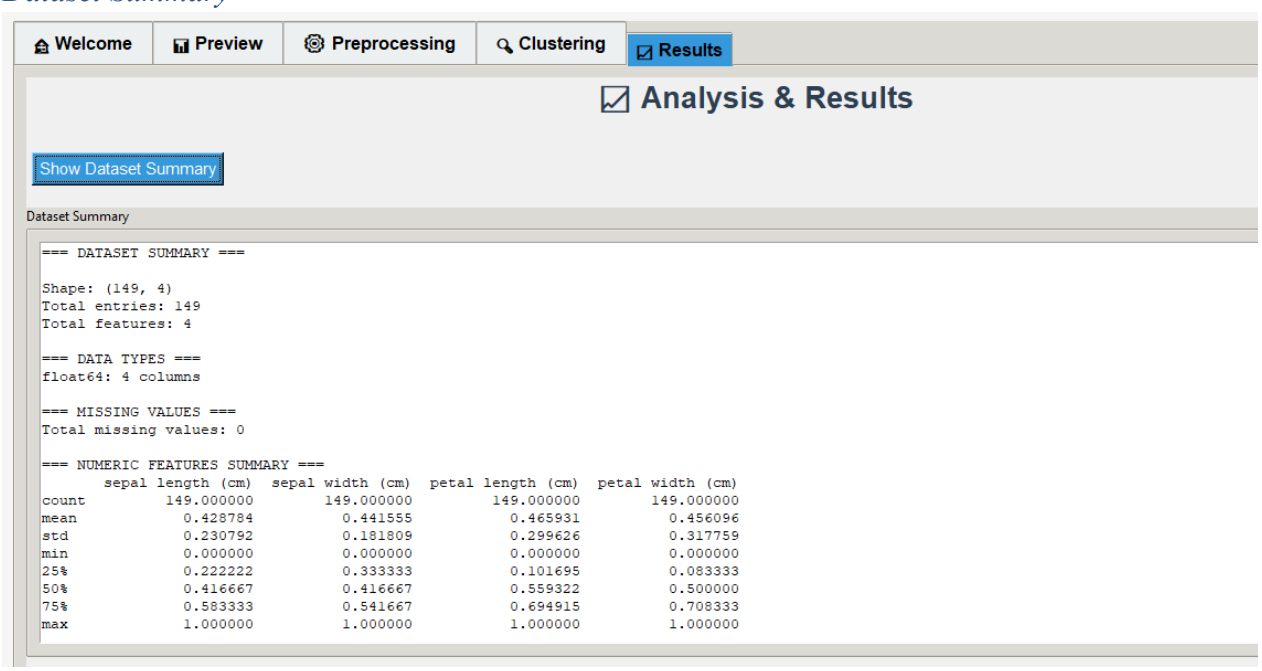
Algorithm	Accuracy	Target Column
KNN	0.633	cp_data
Decision Tree	0.667	cp_data
Naive Bayes	0.278	cp_data
SVM	0.733	cp_data

Classification visualisations



Results Tab

Dataset Summary



VI. Conclusion

My DMI is a comprehensive data mining platform that successfully combines clustering and classification capabilities. The application provides an intuitive interface for data

preprocessing, algorithm comparison, and result visualization, making it both an effective educational tool and a practical solution for data analysis tasks.